

Role of magic angle

Research has proven that magic angles are real. Physicists are trying to apply the twistronics approach to other configurations of graphene. Researchers have been experimenting with twisting two layers of graphene and have found evidence of superconductivity and correlated physics. Others are stacking up three or more layers of graphene in the hopes of gaining superconductivity at other magic angles, or perhaps even when they are aligned. Physicists point out that higher layer stacking may lead to superconductivity temperature to climb.

Other magic angles may play a role, too. Some groups are squeezing sheets of graphene more tightly together to increase the magic angle, making it easier to achieve, and other materials are also coming into the fold of twistronics.

Semiconductors and transitional metals can be deposited in twisted layers, and are seen as good candidates for correlated physics—perhaps, better than twisted bilayer graphene. Researchers are thinking of hundreds of materials that can be manipulated this way in the hope of boosting correlated effects in magic angle twisted bilayer graphene devices by literally smoothing out the wrinkles in their fabrication.

Because there is no chemical bonding between the two layers, and because slightly offset layers try to settle into alignment, forcing them to hold a magic angle twist creates stresses that lead to sub-microscopic hills, valleys and bends. Those local distortions mean that some regions of the device might be within the magic range of twist angles, while other regions are not.

Development towards twistronics

In 2007, National University of Singapore physicist Antonio Castro Neto hypothesised that pressing two misaligned graphene sheets together might yield new electrical properties, and separately proposed that graphene might offer a route to superconductivity, but he did not combine the two ideas.

The theoretical basis of twistronics was further developed by Allan MacDonald in 2011 by predicting that a specific magic angle would radically change the amount of energy a free electron would require to tunnel between two graphene sheets. This hypothesis was confirmed by Pablo Jarillo-Herrero and his student Yuan Cao of MIT and colleagues from Harvard and National Institute for Materials Science in Tsukuba, Japan. They verified in 2018 that superconductivity existed in bilayer graphene where one layer was rotated by an angle of 1.1 relative to the other, forming a moiré pattern, at a temperature of 1.7K.

They created two bilayer devices that acted as an insulator instead of a conductor under a magnetic field; increased field strength turned the second device into a superconductor. Because the only variable is field strength, studying superconductivity in bilayer graphene is faster and more convenient than working with multi-element cuprate crystals.

Experiments have also been done using combinations of graphene layers with other materials that form heterostructures in the form of atomically-thin sheets that are held together by the weak Van der Waals force. A study in July 2019 found that, with addition of a boron-nitride lattice between two graphene sheets, unique orbital ferromagnetic effects were produced at a 1.17 angle, which could be used to implement memory in quantum computers.

Applications

Twistronics is an approach to device engineering in which twist angle between layers of 2D materials such as graphene affects their electronic, optical and mechanical properties. Future applications can be summarised as:

- It will help in the development of new kinds of switching technologies, for example, tunable sensors and tunable electro-mechanical and electro-optical-based devices.
- Though single layers of graphene do not superconduct on their own, they exhibit superconductivity in twisted bilayer resulting from the interaction between layers. This will

lead to various applications of superconducting graphene.

- Researchers believe that it might be possible to introduce, and turn on and off, similar effects, such as spontaneous magnetic ordering and topologically-protected band gaps, with controlled twisting. Results using engineered artificial excitonic crystals of Van der Waals heterostructures highlight the possibility of their applications in nanophotonics and quantum information.

- The work has fundamental significance, providing an attractive platform from which to explore and control excited states of matter, such as topological excitons and a correlated exciton Hubbard model, in transition metal dichalcogenides.

Future scope

Misaligned stacks of 2D materials on twisting are exhibiting different curious properties, and twistronics is gaining steam and interest of many research teams around the world. It is expanding to include other types of materials beyond graphene. The result does not mean getting superconductivity at ambient temperature and to use it in everyday applications.

The property is seen at the very nano-metric level, but what makes it interesting for physicists is that it can open a path to understand superconductivity, and that understanding may open up practical approaches towards usable ambient temperature superconductivity. The ability to tune properties of a layered material by varying the twist angle provides the possibility for a single material to perform a variety of functions instead of using many materials for different functions.

Presently, electronic circuits are built from a finite number of components including metallic conductors, insulators, semiconductors and magnetic materials, and this process requires integration of a variety of different materials, posing significant engineering challenges. But single materials exhibiting change of properties with twisting can be used to realise each of these components, enabling significant new engineering opportunities. **EFY**